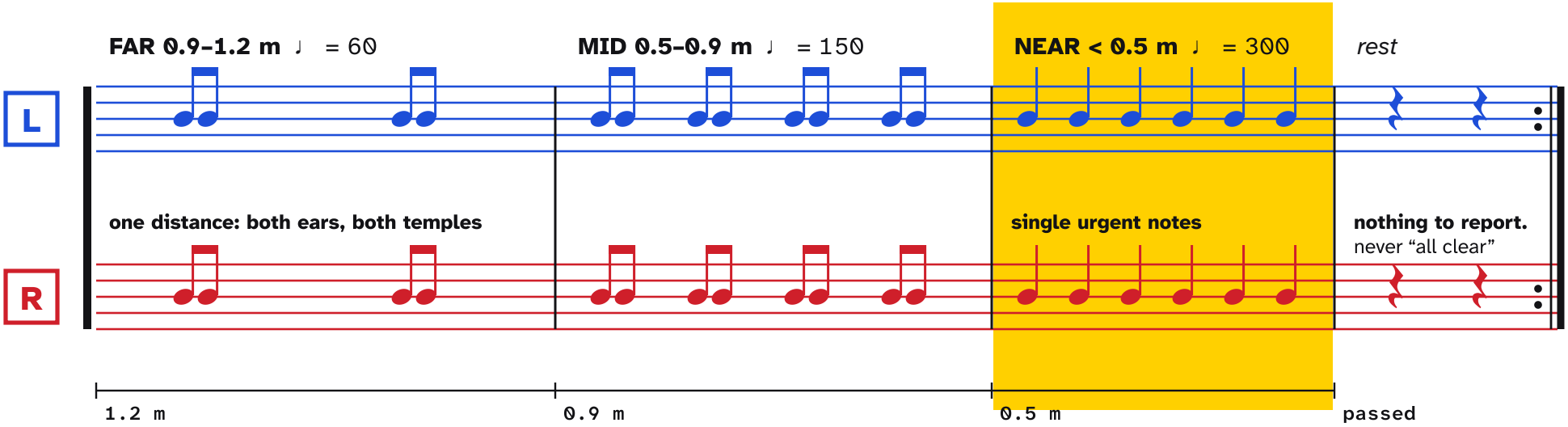
# Determine

**Dumb glasses. Your phone does the thinking. A warning you can hear or feel.**



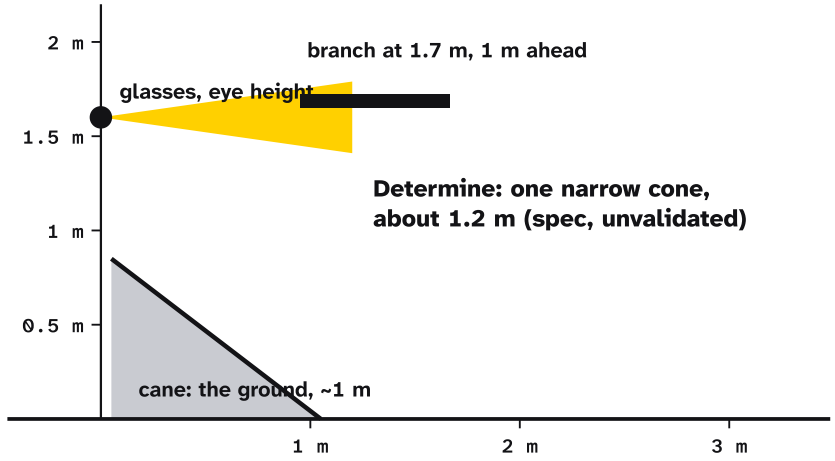

## Cars beep when we can't see behind us. People who are blind deserve a head-level warning too.

**Proposed design · HackTX 26**
Not built or tested yet
UT Austin, 24–25 Oct 2026
For judges, Meta and Annapurna Labs
Full architecture contract on request

# A A cane finds the ground. The branch is at head height.

## About 1 in 8 blind travellers hit their head at least once a month.

In a survey of 307 blind travellers, 13% had a head-level accident at least monthly. 86% of head-level accidents happened outdoors, mostly on tree branches, and 23% had medical consequences.[2] A long cane sweeps the ground ahead of you. Research classes it as a primary aid,[1] and it is excellent at curbs and steps. It cannot feel a branch at eye level, an open cabinet door or a truck mirror.




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### The long cane “does not protect users from obstacles at the upper body and head levels.”

Pittet et al., *Scientific Reports*, 2026[1]

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### Maya

ILLUSTRATIVE TRAVELLER

Walks to class with a long cane. Her left ear is weaker than her right.

### Keeps her cane

Determine adds a head-level layer. It never replaces the cane or a guide dog, and it never says the path is clear.

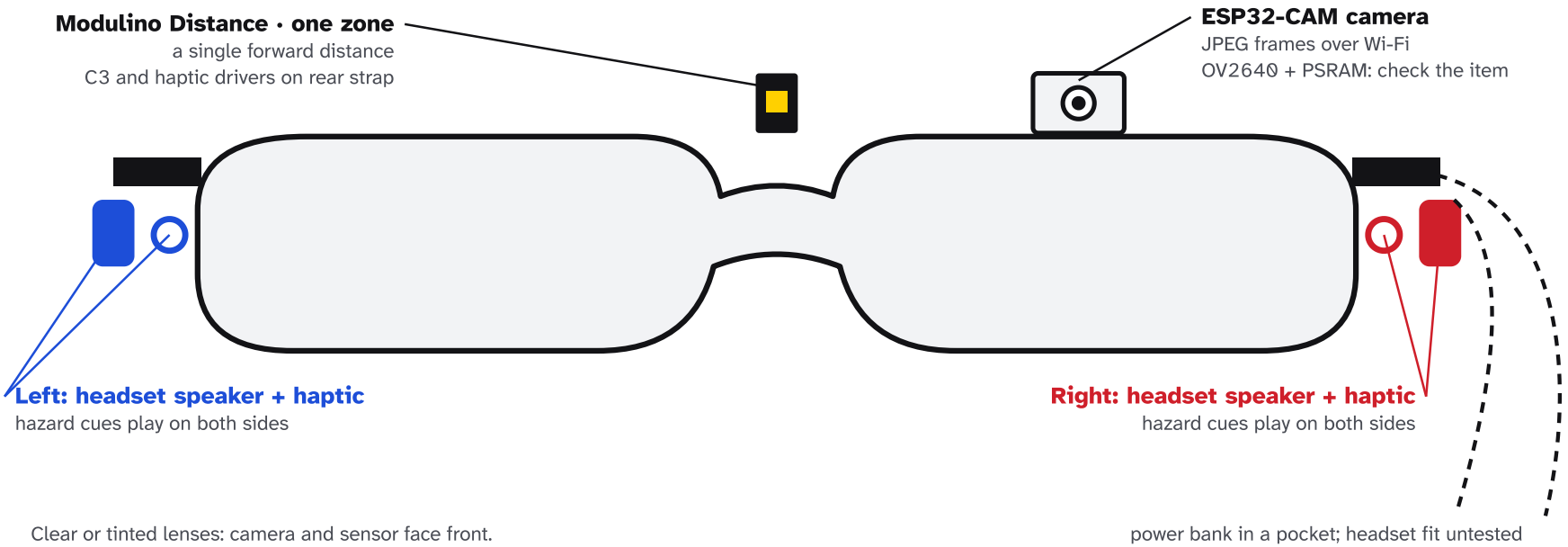
### Wants a warning, not a narrator

In a 2026 trial, every participant preferred short, precise output (“person to your left at 11 o'clock, 3 metres”) to generative descriptions.[1]

### Learns it in minutes

4 of 13 trial users struggled with a rival's spatialised sounds.[1] Determine uses a small cue set, taught at setup, and confirms each warning twice: headset sound and temple buzz.

# B Dumb glasses. Your phone does the thinking.




### Sense · small front, boards behind

The ESP32-CAM sends JPEG frames; one Modulino Distance reports one forward distance through the rear C3. Two Wi-Fi streams reach the phone, which makes every hazard decision.

### Decide · on the phone

The phone aligns both streams by timestamp, and fixed rules turn the forward distance into *one* band: far, mid or near. Camera frames never enter that decision.

### Play · software path

Phone cue audio currently uses a Bluetooth headset; C3 haptics buzz both temples. The separate headset is not approved for the build. Integrated bone-conduction output remains unresolved.

**One forward zone.** It measures how far ahead, whatever the object; it cannot say left or right, and nothing from the camera is attached to it. Spec reach: 1.2 m indoors on a white target, not a validated warning distance.[7]

**Other glasses.** Meta's Wearables DAT 1.0 can stream a supported pair's camera to a phone;[9] Determine's frame input could accept it. Claim only if built at HackTX.

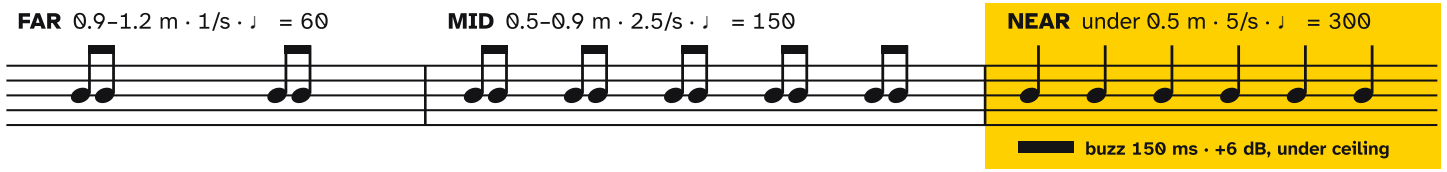
**Two boards, one hotspot.** The C3 has no camera support,[10] so a separate ESP32-CAM supplies JPEG; the second camera is a spare, not stereo. The phone drops stale frames and samples. No inter-board wire.

**Weight off the nose.** Only the camera and sensor face forward; the C3 and drivers ride a rear strap, the power bank a pocket.[6] Fit with the headset is untested.

# C One forward distance. A warning you can hear or feel.

### How close → tempo

The car-sensor model everyone already knows. Never above 6 notes a second.[3] Bands are proposed and tuned by measurement.



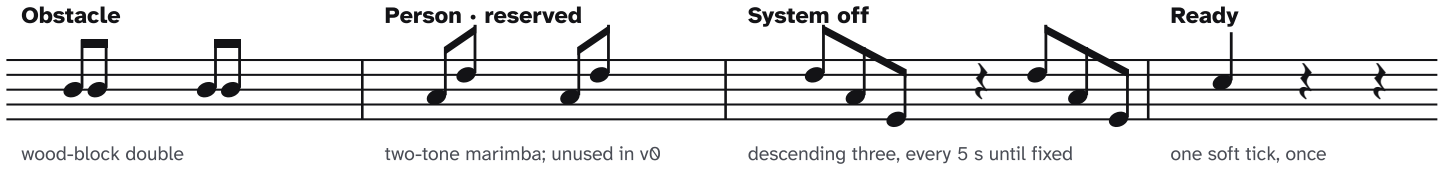

### Where → both, always

One forward zone cannot tell left from right, so every hazard plays on both ears and buzzes both temples. Left and right exist only for calibration tones.




### What → timbre

Timbre and rhythm are the strongest cues for telling sounds apart.[3]




### Words → optional

Sped-up spoken words (spearcons) after the beep, never instead of it. Never a side, never an object name.[4]

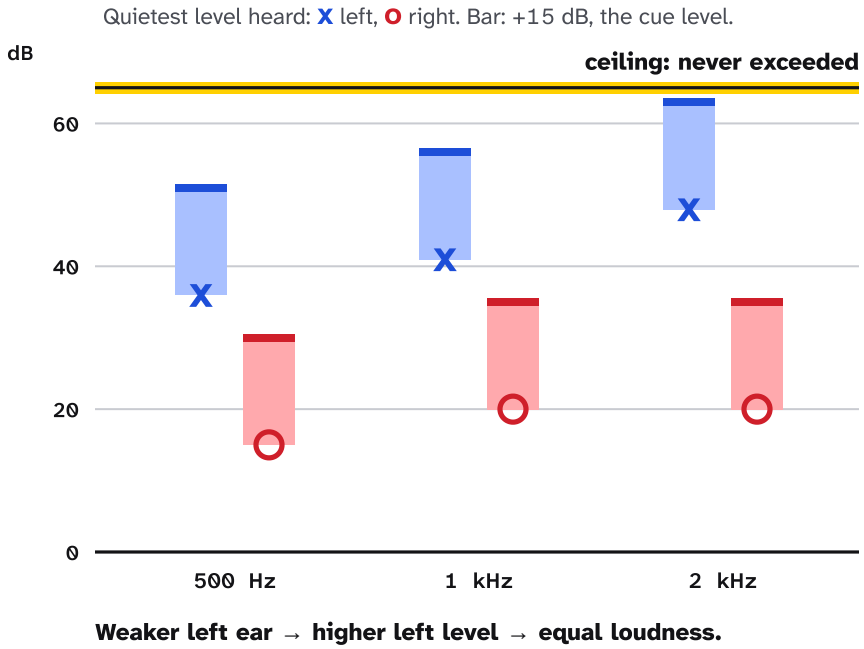
**"obstacle" "near" "system off" "about 1 metre"**



## There is no all-clear sound. Silence is a rest, not the end of the piece.

Silence would mean “nothing to report from a working system”, never “path clear”. If the camera, the distance sensor, either board's link or the headset route fails, the design targets a descending **system off** motif within one second on a surviving output: headset audio, temple haptics or phone vibration. If the phone itself dies, only the board's temple buzz remains, with no sound. Timing and audibility are untested. Tempo, motif and haptics carry urgency too.

# D Equally loud in both ears, even when one is weaker.




**Weaker left ear → higher left level → equal loudness.**

Uses the audiology convention (**X** left, **O** right) but it is not a hearing test and never shows results as one. Values are illustrative.

## Two to three minutes at setup, a 20-second check each time you put them on.

**1 Fit**
Seat the headset. The app requires the bone-conduction route to be active and notes which side feels weaker.

**2 Find each side's quietest audible level**
An adaptive staircase, tap “heard” or “not heard”, in the three bands the cues use.

**3 Set the level**
10–20 dB above that threshold, your choice, clamped to the ceiling.[3]

**4 Balance**
The real obstacle cue alternates left and right. Nudge until both feel the same.

**5 Verify**
Name the side of five random cues. Four right keeps stereo; otherwise switch mode.

### Stereo

Each ear gets its own calibrated level in the headset. Every hazard cue still plays on both, and both temples confirm it.

### One deaf ear

Bone-conducted sound crosses the skull losing only about 0–10 dB,[5] so left/right audio can't carry direction. Audio goes the same to both sides; hazard cues never carried a side anyway, and the temple buzz confirms each one.

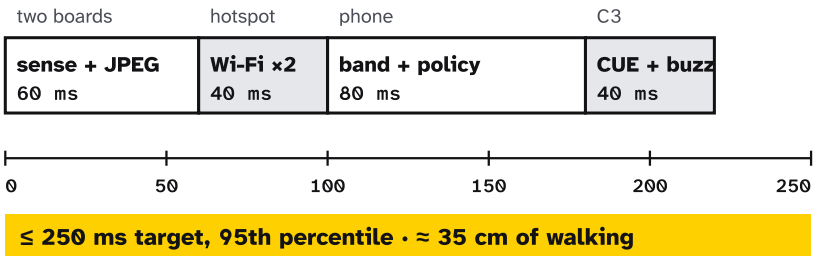
### Haptic only

For anyone who prefers it: the phone plays no cue audio and the temples carry the warning. No step ever passes the audio ceiling. Haptic strength can be lowered, never to zero, so the last local failure buzz stays possible.

# E Two loops. One can't wait. One can think.

Warnings never touch the network. Scene description is a question Maya chooses to ask; the design target is AWS's Annapurna silicon: a Graviton API and a small vision-language model on Inferentia2. Trainium2 is the documented alternative if the Inf2 path fails.[11,13-15]

## Reflex PHONE · OFFLINE · ALWAYS ON




**≤ 250 ms target, 95th percentile · ≈ 35 cm of walking**

Target ≤ 250 ms from range sample to temple buzz at the 95th percentile, shown live on the judges' screen. A goal until measured, and nothing is measured yet; the shared hotspot carrying two streams is the least certain part. Headset sound follows 100–200 ms later over Bluetooth, so the buzz carries urgency. At 1.4 m/s, 250 ms is about 35 cm and the sensor reaches about 1.2 m: a slow, staged near-field demo.

Three tiers of silicon, placed by latency, power and failure budget: two ESP32-class boards on the glasses (a C3 for one-zone range and haptics[6]; a classic-ESP32 camera[10]), the phone for every decision and every sound, and Annapurna silicon for the one job that can wait seconds.

## Describe ON REQUEST · ANNAPURNA SILICON

| | |
|---|---|
| ASK | Long-press: **"what's ahead?"** Never automatic. |
| REDACT | The phone would blur faces and sensitive details before upload; imperfect redaction is a privacy risk, so upload requires the user's choice. |
| GRAVITON | A small arm64 Lambda API (Graviton2) would hold the prompt: two sentences, surroundings first, no guesses about people.[17] |
| INFERENTIA2 | SmolVLM on one inf2.xlarge ($0.76/h) using Hugging Face Optimum Neuron. The model family is listed for Inf2; this deployment is still untested by us.[14,15] |
| SPEAK | The phone would say **"may be wrong"**, then read the result. Any mid- or near-range hazard interrupts it. |

Kept short on purpose: trial participants preferred concise, precise output to generative description.[1] End-to-end time is unmeasured; we plan for seconds, not milliseconds.

### Maya asks

"What's ahead?" The user, not the model, chooses when description is useful. A fast hazard beep would interrupt this answer.

### "May be wrong. A doorway is open on your right; a person is farther ahead near the wall."

ILLUSTRATIVE COPY · NOT MODEL OUTPUT OR A NAVIGATION INSTRUCTION

Describe adds context after the urgent Reflex loop. It never reassures the user that an area is safe, and cloud failure never silences the local cue. Time to response is not measured.

### Inferentia2 · target

SmolVLM via Hugging Face Optimum Neuron on inf2.xlarge. Supported model family, not a validated Determine deployment.[14,15]

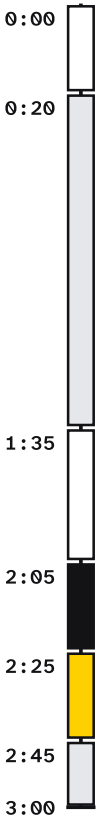
### Trainium2 · alternative

AWS Neuron documents Qwen2-VL-7B on one chip; fit on a smaller trn2.3xlarge is untested.[12,13,16]

### Bedrock · fallback

Nova Lite can handle images, but Bedrock does not guarantee the silicon. New accounts start with Inf/Trn quota 0.[18,19]

# F Three minutes, one course, one pulled plug.



**0:00 Hook** 20 s
The parking-sensor line, moved from the bumper to the brow.

**0:20 Live course** 75 s
A spotter walks the wearer slowly head-on toward a light foam board, then past a head-height bar. The judges' screen shows both streams' age, the one forward distance and its status, the band and measured latency. No side is claimed.

**1:35 One-ear mode** 30 s
Switch to a single-sided profile: headset audio goes mono, the temples keep buzzing for every hazard.

**2:05 Pull the plug** 20 s
Unplug the camera board. "System off" plays; reconnect and wait for ready. Never silence.

**2:25 Ask** 20 s
After recovery, "What's ahead?" If cloud is ready, send a redacted frame for a two-sentence description.

**2:45 Close** 15 s
What we measured, what it can't see, what's next.

**3:00**

## The parts, one store

| | |
|---|---|
| MODULINO DISTANCE | SKU 992826: 2 in Houston, $11.99. One forward zone; not lateral depth.[7] |
| AUDIO UNRESOLVED | Separate OpenRun not approved; old link is support-only. No suitable stocked discrete bone transducer identified.[8] |
| BOARDS | C3 3-pack 069005: 3 packs, $15.59 (one used, two spare). Camera pair 221739: 6 pairs, $17.99 (one spare, not stereo).[6,10] |
| HAPTIC SHORTAGE | Drivers: 4/$7.99; motors: 1/$1.99, need 2; mux: 2/$10.99; FTDI: 2/$7.99. LM3671 sold out. TPS62827 and Vibro are candidates, not drop-ins. |
| POWER + FIT | Stock-checked supply candidates in Architecture §12.2. Dedicated mounts require fabrication; fit untested. |

Houston website snapshot: 24 Oct, ~1:40 PM CDT, checked in Zen. Not reserved; prices may change. No complete-build total or approved integrated audio. Micro Center only.

## Built at HackTX, by four people

The official rule: "All work must be done at the event."[20] Parts are bought ahead; firmware, app and frame are built on site.

**Glasses firmware**
Two boards: C3 runs one-zone range and temple haptics; ESP32-CAM streams JPEG

**Phone perception**
Stream alignment, distance bands, hazard rules, judge screen

**Sound and calibration**
Phone-side cues to the headset, balance flow, one-ear mode

**Describe and demo**
Graviton API, Neuron model, the pitch

Demo branches: no accelerator access → skip the scene answer; camera won't stream → phone camera on a headband, labelled a software fallback. No prerecorded result or fabricated latency.

# G What it can't see, said out loud.

| | |
|---|---|
| **Glass doors and windows** | LIGHT PASSES THROUGH |
| **Thin wires and ropes** | TOO THIN FOR ONE ZONE |
| **Drop-offs, curbs, stairs going down** | THE CANE'S JOB |
| **Anything outside one narrow cone** | 18° DIAGONAL |
| **Which side, or what it is** | ONE ZONE, NO LABEL |
| **Fast-moving vehicles** | NOT A TRAFFIC AID |
| **Glare, darkness, rain, sun, gray surfaces** | DEGRADES, ANNOUNCED |
| **Anything beyond about 1.2 m** | VENDOR SPEC, UNVALIDATED |

Spoken to every new user during setup. Demos run on a staged indoor course with a sighted spotter.

**Fail loud.**
Design target: announce “system off” within one second through a working output (headset, temple buzz or phone vibration). If the phone dies, only the board's buzz remains. Never call silence a clear path. Not yet tested.

**Unknown is not clear.**
No target or an unreadable sample never produces a reassuring sound.

**The cane stays.**
Determine is a secondary aid, on top of a cane or guide dog.

**Claim only what we measured.**
Every threshold and latency here is a target until tested with blind and low-vision people, with an orientation and mobility specialist.

**Sources (15–24 Oct 2026; Houston stock snapshot on page 7)**


1. Pittet et al., secondary travel-aid trial, 2026
2. Manduchi & Kurniawan, mobility accidents, 2011
3. Brewster et al., earcon guidelines, 1995
4. Walker et al., spearcons, 2013
5. Stenfelt, sound crossing the skull, 2012
6. Seeed, XIAO ESP32-C3 wiki: 400 KB SRAM, one I2C, no camera
7. Arduino, Modulino Distance; ST, VL53L4CD datasheet: 1200 mm indoors, 18° FOV; Micro Center listing
8. OpenRun: old link redirects to support, not approved; Humbird: no longer carried; DRV2605L
9. Meta, Wearables DAT and Mock Device Kit
10. Espressif, esp32-camera driver: ESP32/S2/S3; PSRAM above-CIF JPEG
11. About Amazon, Annapurna Labs
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13. AWS Neuron, Qwen2-VL tutorial
14. Hugging Face, Optimum Neuron models
15. AWS, EC2 Inf2 instances
16. AWS, EC2 Trn2 instances
17. AWS, Lambda arm64 / Graviton2
18. AWS, Bedrock Nova Lite model card
19. AWS, EC2 accelerator quotas
20. Freetail Hackers, HackTX 26 FAQ
21. HackTX, Annapurna Labs platinum partner post